

5 The Qualitative-Graphic Approach

The several cases of nonlinear differential equations previously discussed (exact differential equations, separable-variable equations, and Bernoulli equations) have all been solved *quantitatively*. That is, we have in every case sought and found a time path $y(t)$ which, for each value of t , tells the specific corresponding value of the variable y .

At times, we may not be able to find a quantitative solution from a given differential equation. Yet, in such cases, it may nonetheless be possible to ascertain the *qualitative* properties of the time path—primarily, whether $y(t)$ converges—by directly observing the differential equation itself or by analyzing its graph. Even when quantitative solutions are available, moreover, we may still employ the techniques of qualitative analysis if the qualitative aspect of the time path is our principal or exclusive concern.

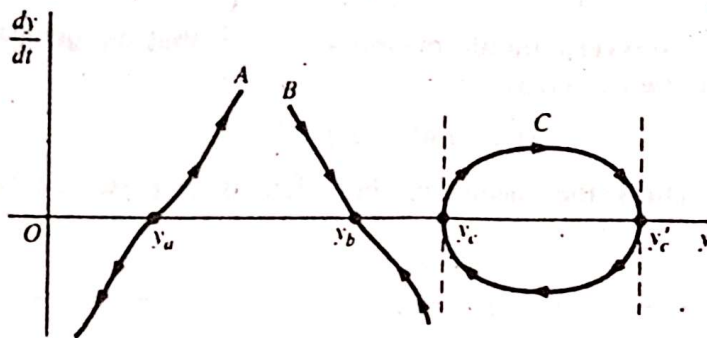
The Phase Diagram

Given a first-order differential equation in the general form

$$\frac{dy}{dt} = f(y)$$

either linear or nonlinear in the variable y , we can plot dy/dt against y as in Fig. 15.3. Such a geometric representation, feasible whenever dy/dt is a function of y alone, is called a *phase diagram*, and the graph representing the function f , a *phase line*. (A differential equation of this form—in which the time variable t does not appear as a separate argument of

FIGURE 15.3



the function f —is said to be an *autonomous* differential equation.) Once a phase line is known, its configuration will impart significant qualitative information regarding the time path $y(t)$. The clue to this lies in the following two general remarks:

1. Anywhere *above* the horizontal axis (where $dy/dt > 0$), y must be **increasing over time** and, as far as the y axis is concerned, must be moving from left to right. By analogous reasoning, any point *below* the horizontal axis must be associated with a **leftward movement** in the variable y , because the negativity of dy/dt means that y decreases over time. These directional tendencies explain why the arrowheads on the illustrative phase lines in Fig. 15.3 are drawn as they are. Above the horizontal axis, the arrows are uniformly pointed toward the right—toward the northeast or southeast or due east, as the case may be. The opposite is true below the y axis. Moreover, these results are independent of the algebraic sign of y ; even if phase line A (or any other) is transplanted to the left of the vertical axis, the direction of the arrows will not be affected.
2. An equilibrium level of y —in the intertemporal sense of the term—if it exists, can occur only on the horizontal axis, where $dy/dt = 0$ (y stationary over time). To find an equilibrium, therefore, it is necessary only to consider the intersection of the phase line with the y axis.[†] To test the dynamic stability of equilibrium, on the other hand, we should also check whether, regardless of the initial position of y , the phase line will always guide it toward the equilibrium position at the said intersection.

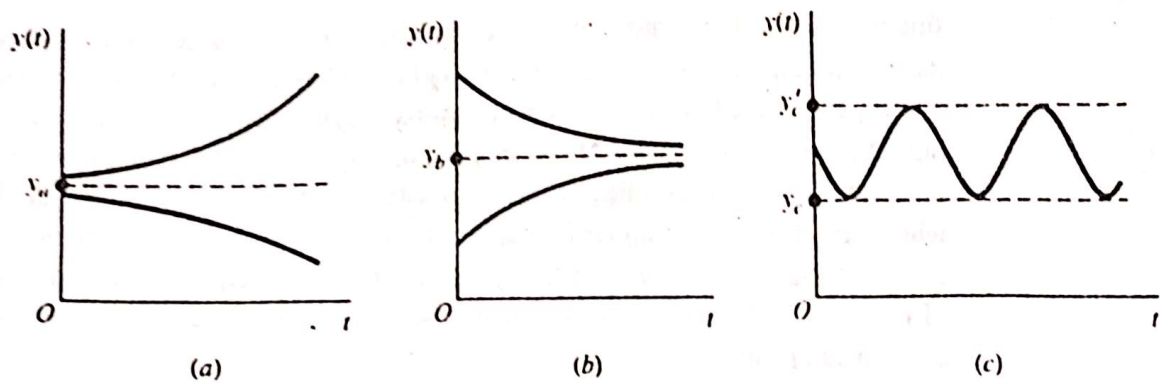
Types of Time Path

On the basis of the preceding general remarks, we may observe three different types of time path from the illustrative phase lines in Fig. 15.3.

Phase line A has an equilibrium at point y_a ; but *above* as well as *below* that point, the arrowheads consistently lead away from equilibrium. Thus, although equilibrium can be attained if it happens that $y(0) = y_a$, the more usual case of $y(0) \neq y_a$ will result in y being ever-increasing [if $y(0) > y_a$] or ever-decreasing [if $y(0) < y_a$]. Besides, in this case the deviation of y from y_a tends to grow at an increasing pace because, as we follow the arrowheads on the phase line, we deviate farther from the y axis, thereby encountering ever-increasing numerical values of dy/dt as well. The time path $y(t)$ implied by phase line A can therefore be represented by the curves shown in Fig. 15.4a, where y is plotted against t (rather than dy/dt against y). The equilibrium y_a is dynamically unstable.

[†] However, not all intersections represent equilibrium positions. We shall see this when we discuss phase line C in Fig. 15.3.

FIGURE 15.4



In contrast, phase line *B* implies a stable equilibrium at y_b . If $y(0) = y_b$, equilibrium prevails at once. But the important feature of phase line *B* is that, even if $y(0) \neq y_b$, the movement along the phase line will guide y toward the level of y_b . The time path $y(t)$ corresponding to this type of phase line should therefore be of the form shown in Fig. 15.4b, which is reminiscent of the dynamic market model.

The preceding discussion suggests that, in general, it is the slope of the phase line at its intersection point which holds the key to the dynamic stability of equilibrium or the convergence of the time path. A (finite) *positive* slope, such as at point y_a , makes for dynamic *instability*; whereas a (finite) *negative* slope, such as at y_b , implies dynamic *stability*.

This generalization can help us to draw qualitative inferences about given differential equations without even plotting their phase lines. Take the linear differential equation in (15.4), for instance:

$$\frac{dy}{dt} + ay = b \quad \text{or} \quad \frac{dy}{dt} = -ay + b$$

Since the phase line will obviously have the (constant) slope $-a$, here assumed nonzero, we may immediately infer (without drawing the line) that

$$a \gtrless 0 \quad \Leftrightarrow \quad y(t) \left\{ \begin{array}{l} \text{converges to} \\ \text{diverges from} \end{array} \right\} \text{equilibrium}$$

As we may expect, this result coincides perfectly with what the quantitative solution of this equation tells us:

$$y(t) = \left[y(0) - \frac{b}{a} \right] e^{-at} + \frac{b}{a} \quad [\text{from (15.5')}]$$

We have learned that, starting from a nonequilibrium position, the convergence of $y(t)$ hinges on the prospect that $e^{-at} \rightarrow 0$ as $t \rightarrow \infty$. This can happen if and only if $a > 0$; if $a < 0$, then $e^{-at} \rightarrow \infty$ as $t \rightarrow \infty$, and $y(t)$ cannot converge. Thus, our conclusion is one and the same, whether it is arrived at quantitatively or qualitatively.

It remains to discuss phase line *C*, which, being a closed loop sitting across the horizontal axis, does not qualify as a *function* but shows instead a *relation* between dy/dt and y .[†] The interesting new element that emerges in this case is the possibility of a periodically fluctuating time path. The way that phase line *C* is drawn, we shall find y fluctuating between the two values y_c and y_c' in a perpetual motion. In order to generate the periodic

[†] This can arise from a second-degree differential equation $(dy/dt)^2 = f(y)$.

fluctuation, the loop must, of course, straddle the horizontal axis in such a manner that dy/dt can alternately be positive and negative. Besides, at the two intersection points y_c and y'_c , the phase line should have an infinite slope; otherwise the intersection will resemble either y_a or y_b , neither of which permits a continual flow of arrowheads. The type of time path $y(t)$ corresponding to this looped phase line is illustrated in Fig. 15.4c. Note that whenever $y(t)$ hits the upper bound y'_c or the lower bound y_c , we have $dy/dt = 0$ (local extrema); but these values certainly do not represent equilibrium values of y . In terms of Fig. 15.3, this means that not all intersections between a phase line and the y axis are equilibrium positions.

In sum, for the study of the dynamic stability of equilibrium (or the convergence of the time path), one has the alternative either of finding the time path itself or else of simply drawing the inference from its phase line. We shall illustrate the application of the latter approach with the Solow growth model. Henceforth, we shall denote the intertemporal equilibrium value of y by $\bar{y}$, as distinct from y^* .